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|  | <b>RV College of Engineering®</b><br><b>Department of Computer Science and Engineering</b><br><b>CIE - II: Test Paper</b> |                             |                      |
| <b>Course &amp; Code</b>                                                          | <b>Deep Learning</b><br><b>(CS374TFA)</b>                                                                                 |                             | <b>Semester: VII</b> |
| <b>Date: 16/12/2025</b>                                                           | <b>Duration:</b> 90 minutes                                                                                               | <b>Max.Marks :</b> 50 Marks | <b>Staff : RJ</b>    |
| <b>USN :</b>                                                                      | <b>Name :</b>                                                                                                             |                             |                      |

**NOTE: Answer all the questions**

**Part -A**

| Sl.no. | Questions                                                                                                                                                                                                                                                                                               | Marks | BT | CO  |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|-----|
| 1.     | What is teacher forcing in recurrent neural networks? Why is it used during training?                                                                                                                                                                                                                   | 2     | L2 | CO1 |
| 2.     | Apply a real-world scenario where a CNN and an RNN can be combined, explaining how this combination benefits the task.                                                                                                                                                                                  | 2     | L3 | CO4 |
| 3.     | Compare tiled convolution with locally connected layers in terms of parameter sharing and memory requirements.                                                                                                                                                                                          | 2     | L4 | CO2 |
| 4.     | A researcher wants to classify musical notes using a CNN. The audio waveform is first converted into a spectrogram where rows represent frequencies and columns represent time. What type of data format is this (1-D, 2-D, or 3-D), and why is a convolutional model suitable for this representation? | 2     | L3 | CO2 |
| 5.     | Analyze the challenges faced when training deep recurrent networks using standard gradient-based optimization methods.                                                                                                                                                                                  | 2     | L4 | CO1 |

**Part B**

| Sl.no. | Questions                                                                                | Marks | BT | CO  |
|--------|------------------------------------------------------------------------------------------|-------|----|-----|
| 1.     | With a block diagram, explain the working principle of LSTM with all relevant equations. | 10    | L2 | CO3 |
| 2.     | Briefly explain the variants of Convolution Neural Networks.                             | 10    | L2 | CO2 |
| 3.     | Describe and compare the major design patterns of recurrent neural networks.             | 10    | L2 | CO3 |

|    |                                                                                                                                                                                                                                                                                           |    |    |     |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| 4. | Explain the Encoder–Decoder Sequence-to-Sequence RNN architecture, describing how the encoder forms the context vector and how the decoder generates the output sequence. Also discuss the limitation of using a fixed-size context vector and how attention addresses this issue.        | 10 | L3 | CO3 |
| 5. | A company’s shallow RNN model fails to capture long-term dependencies in complex sequential data. How can deep recurrent network architectures improve performance in this situation? Discuss with reference to deeper transformations, hierarchical hidden layers, and skip connections. | 10 | L3 | CO4 |

#### COURSE OUTCOMES:

|             |                                                                                                                                                                                                                                                                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CO1:</b> | Explore key theoretical concepts like the Universal Approximation Theorem, vanishing / exploding gradients, and optimization methods.                                                                                                                                                                                                                 |
| <b>CO2:</b> | Analyse the fundamental concepts of Deep Learning, and its various architecture learning models, including Neural Networks, backpropagation, gradient descent, and different Network Architectures (feedforward, convolutional, recurrent) Learning tasks for various applications.                                                                   |
| <b>CO3:</b> | Apply the Deep learning model approaches to know the strengths and weaknesses of the architecture by empirical results. Apply appropriate concepts like Recurrent, Recursive Nets and Auto-encoder models to specific real time projects and analyse the Optimization techniques.                                                                     |
| <b>CO4:</b> | Designing and implement a Deep Learning model as part of an experiential learning initiative in teams to solve societal and environmental problems. Ability to fine tune the model parameters to improve performance, explore and understand the ethical implications and societal impact of deploying deep learning systems in real-world scenarios. |

|              | L1 | L2 | L3 | L4 | L5 | L6 | CO1 | CO2 | CO3 | CO4 |
|--------------|----|----|----|----|----|----|-----|-----|-----|-----|
| <b>Marks</b> | -  | 32 | 24 | 4  | -  | -  | 4   | 14  | 30  | 12  |